

DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

# Load the MA_schools dataset
data(MA_schools)
```

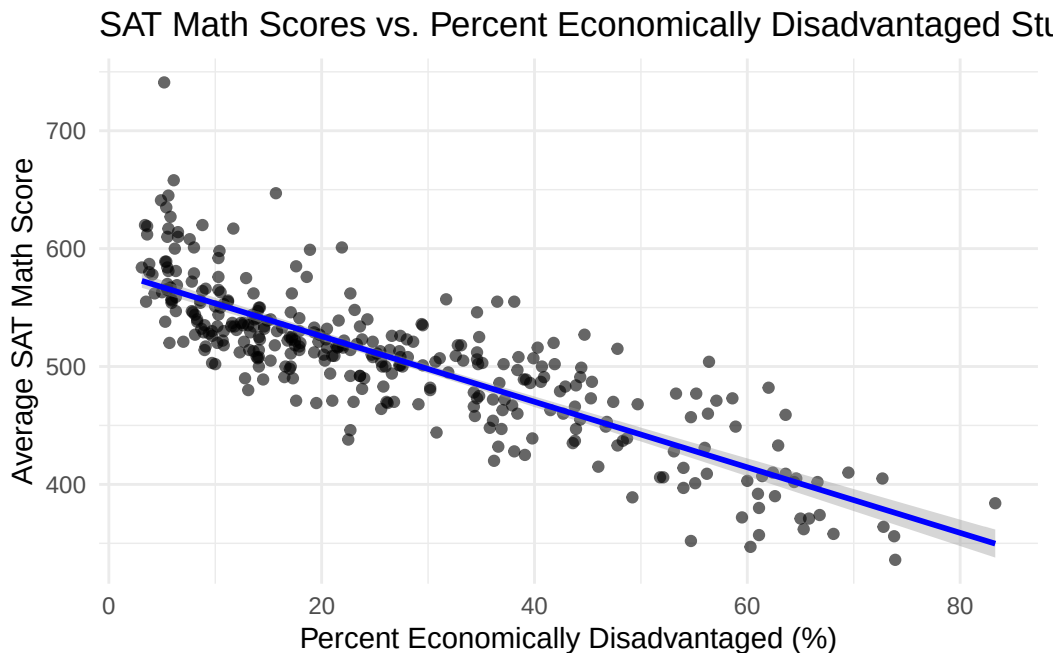
Data Exploration (3 points)

1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

```
ggplot(MA_schools, aes(x = perc_disadvan, y = average_sat_math)) +  
  geom_point(alpha = 0.6) +  
  geom_smooth(method = "lm", se = TRUE, color = "blue") +  
  labs(x = "Percent Economically Disadvantaged (%)",  
       y = "Average SAT Math Score",  
       title = "SAT Math Scores vs. Percent Economically Disadvantaged Students") +  
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'



Answer: Based on the scatterplot, I expect a negative relationship between the percentage of economically disadvantaged students and average SAT math scores. As the percentage of economically disadvantaged students increases, the average SAT math scores tend to decrease, as indicated by the downward-sloping trend in the data.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
model1 <- lm(average_sat_math ~ perc_disadvan, data = MA_schools)
summary(model1)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.74	-21.26	-4.12	18.54	174.17

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	581.2811	3.2668	177.9	<2e-16 ***
perc_disadvan	-2.7798	0.1011	-27.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6953

F-statistic: 756.2 on 1 and 330 DF, p-value: < 2.2e-16

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
sat_effects <- coef(model1)
sat_effects
```

(Intercept)	perc_disadvan
581.281054	-2.779787

Intercept: 581.28

Slope: -2.78

3 (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: For every one percentage point increase in economically disadvantaged students, the average SAT math score is predicted to decrease by approximately 2.78 points, on average. This indicates a negative association between economic disadvantage and SAT math performance, suggesting that schools with higher percentages of economically disadvantaged students tend to have lower average SAT math scores.

Making Predictions (8 points)

1 (4 points)

Use `model1` to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
new_data <- data.frame(perc_disadvan = 50)
predicted_score <- predict(model1, newdata = new_data)
predicted_score
```

```
1
442.2917
```

Predicted average SAT math score: Approximately 442.30

2 (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
prediction_interval <- predict(model1, newdata = new_data,
                              interval = "prediction", level = 0.95)
prediction_interval
```

	fit	lwr	upr
1	442.2917	376.0542	508.5292

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```
model2 <- lm(average_sat_math ~ perc_disadvan + size, data = MA_schools)
summary(model2)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-81.342	-20.802	-3.872	18.211	167.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	588.1898	7.6067	77.325	<2e-16 ***
perc_disadvan	-2.7772	0.1063	-26.120	<2e-16 ***
sizemedium	-11.9134	7.5346	-1.581	0.115
sizelarge	-6.3615	6.9231	-0.919	0.359

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.5 on 328 degrees of freedom

Multiple R-squared: 0.6988, Adjusted R-squared: 0.696

F-statistic: 253.6 on 3 and 328 DF, p-value: < 2.2e-16

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```
r_squared_model1 <- summary(model1)$r.squared
r_squared_model2 <- summary(model2)$r.squared

cat("Model 1 R-squared:", round(r_squared_model1, 4), "\n")
```

Model 1 R-squared: 0.6962

```
cat("Model 2 R-squared:", round(r_squared_model2, 4), "\n")
```

Model 2 R-squared: 0.6988

Answer: Model 2 explains more variance in average SAT math scores. The R-squared value for Model 2 is higher than Model 1, indicating that adding school size as a predictor improves the model's explanatory power.

3 (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```
anova(model1, model2)
```

Analysis of Variance Table

```
Model 1: average_sat_math ~ perc_disadvan
Model 2: average_sat_math ~ perc_disadvan + size
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	330	371191				
2	328	368058	2	3132.6	1.3958	0.2491

Conclusion: Based on the ANOVA test, the p-value (0.2491) is greater than $\alpha = 0.05$. We fail to reject the null hypothesis. This means that adding the size variable does not significantly improve the model at the $\alpha = 0.05$ level. In other words, school size does not provide statistically significant additional explanatory power beyond the percentage of economically disadvantaged students in predicting SAT math scores.